

Create an Ecocolumn!

Introduction:

Students build ecocolumns from stacked 2-liter bottles to model how water filters, or fails to filter, through different materials before reaching a stream. Each column has a "sky" layer (water + simulated pollutants like diluted food coloring, a drop of cooking oil, and a pinch of soil), a soil layer that students design themselves using combinations of gravel, sand, compost, and activated charcoal, and a collection chamber at the bottom.

The challenge is to design the most effective filtration system. Students measure turbidity, color, and pH of their collected water against a control. They test multiple iterations, changing the order or composition of their layers. The best-performing designs get documented and compared across teams.

The connection back to the bigger picture is explicit: a rain garden works because the soil and roots do exactly what their best-performing ecocolumn did. When they understand that at a material level, the rain garden stops being a school project and starts being obvious engineering. It takes about three class periods. Materials cost under \$15 per team.

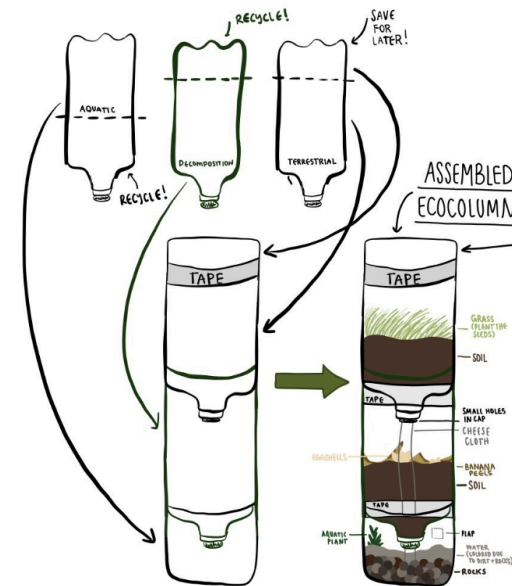
Materials (Teams of 3-4 students):

Material	Quantity
Empty 2-liter bottles	3
Gravel	2 cups
Cheese cloth	1 pc
Compost/soil	2 cups
Compost materials (Seeds, banana peels, eggshells)	As needed
Mixed grass seeds	As needed
Aquatic plant (e.g. elodea)	1
Treated water	As needed
Utility knife and scissors	1

Instructions:

Building the EcoColumn

1. Create the Aquatic Chamber
 - a. Take the first 2-liter bottle and using scissors or a utility knife, carefully cut the bottom in half around the middle
 - b. Keep the bottom half of the bottle. This will become your aquatic chamber which represents a pond, wetland, or stream ecosystem
 - c. Fill the chamber with a layer of gravel, treated water until the chamber is approximately 2/3s full, and one aquatic plant
2. Create the Connector Section
 - a. Take the second bottle
 - b. Cut off both the top and bottom portions, leaving the cylindrical middle section (Save the pieces you cut for later)
 - c. Place the cylinder on top of the aquatic chamber and secure with tape
3. Install the Drainage Wick
 - a. Locate the bottle cap with a pre-drilled hole and thread a strip of cheese cloth through the hole
 - b. Pull enough cloth so that it hangs several inches on both sides; the cheese cloth should act as a wick, allowing water to move downward through the ecocolumn while keeping soil in place
4. Building the Decomposition Chamber
 - a. Take the third bottle and cut off the bottom section
 - b. Screw the prepared cap with the cheesecloth wick onto the bottle
 - c. Turn the bottle upside down so the neck points downward and place it securely on top of the connector section
5. Build the Terrestrial Chamber
 - a. Repeat the same process used for the decomposition chamber with another prepared bottle section
 - b. Install a bottle cap with a cheesecloth wick
 - c. Position the chamber above the decomposition layer
 - d. Fill this chamber with potting soil and grass seeds
6. Add the Top Cover
 - a. Retrieve one of the bottle-bottom pieces saved earlier and place it over the top opening like a lid



Testing and Data Collection

1. Preparation for Soil and Nitrate/Nitrite Test
 - a. Mix three grams of soil with three grams of water in a test tube. Each chamber needs a different test tube. You will need two test tubes for this.
 - b. Shake for 30 seconds vertically.
 - c. Place test tubes in a paper cup, oriented vertically. Wait for 10 minutes. The solution needs to be separated before you can start testing.
2. Soil pH Test
 - a. Place one soil pH test strip fully within the solution for 3 seconds. Then, take it out and hold horizontally until the color fully changes. Use a different test strip for each test tube.
 - b. Record the soil pH reading based on the color of the strip. Compare its color to the reference chart provided with the test kit. The color of the strip will indicate the pH level, with lighter colors usually representing more acidic soil (lower pH) and darker, more saturated colors indicating a more basic soil (higher pH). Make sure to check the strip under good lighting to ensure an accurate reading.
3. Nitrate/Nitrite Test
 - a. Dip the nitrate/nitrite combo test strip into the solution vertically so both pads are completely immersed for 2-3 seconds.
 - b. Remove and shake off the excess liquid. Hold horizontally for 30 seconds.
 - c. Compare the results to the color chart. The more saturated the color, the higher the concentration of nitrates/nitrites.
4. Aquarium 7 in 1 Test
 - a. Dip the test strip vertically downwards into the water for 2 seconds.
 - b. Remove and hold the strip horizontally for 60 seconds.
 - c. Compare with the color chart on the bottle. Typically, the more saturated the color, the higher the concentration of the substance.
5. Dissolved Oxygen (DO) Test
 - a. Set the probe to %. Unscrew the protective cover and turn on Lab Quest.
 - b. Insert the plug into Channel 1.
 - c. Insert the DO probe into the solution.
 - d. Select the "play" icon and record for 120 seconds. Write down these results.
 - e. Rinse off the DO probe with fresh water. Dry off with a rag and screw on the probe protector.

Quantitative Data Table

Location	Date

Qualitative Data Table

Location	Test	Date